

Investment Project

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2 The Data

Sources. We use the WRDS Monthly World Indices for Australia, France, Germany, Japan, Switzerland and the United Kingdom. These are total returns including dividends (`mportret`), in local currency. For the US, we use the CRSP value-weighted index. Both were downloaded from the WRDS web interface, from 2002-01 to 2024-12. From FRED, we downloaded three series:

- the exchange rates of the AUD, EUR, JPY, CHF and GBP against the USD (end of month);
- the 3-month interbank rate of each country (monthly averages, annualized);
- the 1-month T-Bill rate, used as the risk-free rate (monthly average, annualized).

The Fama-French 5 factors (Section 9) come from Kenneth French’s website.

Conventions.

- All series are monthly and aligned on the month. WRDS dates are at the end of the month, while FRED dates are on the first day of the month.
- A return indexed by t is earned *during* month t , from the end of $t - 1$ to the end of t .
- Annual rates r_y are converted into monthly rates r_m with $(1 + r_m)^{12} = 1 + r_y$.
- Interest rates are lagged by one month: the rate earned over month t is the one observed in month $t - 1$, when the position is opened. Since FRED rates are monthly averages, using the rate of month t itself would use information that is not yet available.
- More generally, every weight or signal used for month t only uses information available at the end of month $t - 1$. All rolling estimates use a window of 60 months.

Data treatment.

- The US interbank rate is missing for April 2020. We interpolate it linearly; otherwise, this month (the post-Covid rebound) would be dropped from every strategy.
- The Japanese interbank rate starts in April 2002. With the one-month lag, the common sample therefore runs from May 2002 to December 2024 (272 months).
- France and Germany share the same currency and the same 3-month rate (Euribor). The euro is therefore counted only once in the currency strategies.

3 The international diversification strategy (DIV)

- (a) We first compute the dollar returns of the six foreign indices. Let R_t^{loc} be the return of an index in local currency over month t :

$$R_t^{\text{loc}} = \frac{P_t^{\text{loc}} + D_t^{\text{loc}} - P_{t-1}^{\text{loc}}}{P_{t-1}^{\text{loc}}}$$

where P_t^{loc} and D_t^{loc} are the price of the index and the dividends paid, in local currency. Let S_t be the value in USD of one unit of the local currency (for the CHF and JPY, FRED gives the inverse quote, which we invert). A US investor then earns the unhedged return

$$R_t^{\text{unhedged}} = \frac{(P_t^{\text{loc}} + D_t^{\text{loc}}) \times S_t - P_{t-1}^{\text{loc}} \times S_{t-1}}{P_{t-1}^{\text{loc}} \times S_{t-1}} = (R_t^{\text{loc}} + 1) \times \frac{S_t}{S_{t-1}} - 1$$

This return combines the local equity return and the change in value of the foreign currency.

- (b) Let X_t be the excess return in USD of a \$1 investment in the foreign currency. It is obtained by borrowing \$1 at the US 3-month rate r^{US} , converting it into $\frac{1}{S_{t-1}}$ units of local currency, investing them at the local 3-month rate r^{loc} , and converting the proceeds back into dollars after one month:

$$X_t = \frac{S_t}{S_{t-1}} (1 + r_{t-1}^{\text{loc}}) - (1 + r_{t-1}^{US})$$

The rates are indexed by $t - 1$ because they are known when the position is opened. The currency-hedged return removes this pure currency component, which amounts to shorting \$1 of the foreign currency:

$$R_t^{\text{hedged}} = R_t^{\text{unhedged}} - X_t = \frac{S_t}{S_{t-1}} (R_t^{\text{loc}} - r_{t-1}^{\text{loc}}) + r_{t-1}^{US}$$

The hedged return is therefore the local equity excess return plus the US rate. The exchange rate only multiplies the equity excess return, which is a second-order effect.

Table 1 shows the effect of hedging on each index. Hedging strongly reduces the volatility of the Australian, French, German, Swiss and British indices. Japan is the exception: its volatility increases. The yen is a safe-haven currency that tends to appreciate when Japanese equities fall, so keeping the currency exposure acts as a natural hedge for a US investor.

	AUS	FRA	DEU	CHE	JAP	GBR
Unhedged – Mean	11.11%	9.04%	8.85%	8.87%	6.56%	6.01%
Unhedged – Volatility	21.60%	19.90%	21.24%	15.68%	14.63%	16.87%
Unhedged – Sharpe	0.45	0.38	0.35	0.47	0.35	0.27
Hedged – Mean	8.10%	8.49%	8.31%	7.44%	8.62%	5.85%
Hedged – Volatility	13.34%	16.06%	17.42%	13.27%	16.21%	13.08%
Hedged – Sharpe	0.49	0.44	0.39	0.45	0.44	0.33

TABLE 1: Annualized performance of each index in USD, unhedged and currency-hedged (2002-05 to 2024-12)

(c) We build three portfolios of the six foreign indices. In each case, the weights for month $t + 1$ only use returns up to month t :

- **Equally weighted:** we invest $w_{t+1}^i = \frac{1}{N}$ of our wealth in each of the N indices.
- **Risk parity:** each index receives a weight inversely proportional to its volatility over the past 60 months, normalized so that the weights sum to 1:

$$\sigma_t^i = \text{std} \left(R_{t-59}^i, \dots, R_t^i \right), \quad w_{t+1}^i = \frac{1/\sigma_t^i}{\sum_{j=1}^N 1/\sigma_t^j}$$

Less volatile markets thus receive a larger weight. This simple version ignores correlations.

- **Mean-variance:** at each date t , we estimate the mean vector μ_t and the covariance matrix Σ_t of the N indices over the past 60 months. We build the portfolio $R_{t+1}^P = R_{t+1}^f + w_{t+1}^\top (R_{t+1} - R_{t+1}^f \mathbf{1})$ that maximizes the mean-variance utility of an investor with risk aversion γ :

$$\begin{aligned} & \max_{w_{t+1}} \left\{ \mathbb{E}_t[R_{t+1}^P] - \frac{\gamma}{2} \text{Var}_t(R_{t+1}^P) \right\} \\ \iff & \max_{w_{t+1}} \left\{ w_{t+1}^\top (\mu_t - \bar{R}_t^f \mathbf{1}) - \frac{\gamma}{2} w_{t+1}^\top \Sigma_t w_{t+1} \right\} \end{aligned}$$

where $\bar{R}_t^f$ is the average T-Bill return over the same window. The solution is

$$w_{t+1} = \frac{1}{\gamma} \Sigma_t^{-1} \left(\mu_t - \bar{R}_t^f \mathbf{1} \right)$$

The remaining $1 - \sum_i w_{t+1}^i$ is invested in the T-Bill.

The portfolios first contain only the six foreign indices, because they measure what a US investor obtains by investing abroad, and the US market alone serves as the

benchmark. However, a US investor would add foreign equities to US equities rather than replace them. We therefore also build the three portfolios on the US index and the six currency-hedged foreign indices (the US index needs no currency hedge). All portfolios are evaluated over the same period, 2007-05 to 2024-12 (212 months), since the first 60 months are needed for estimation. Table 2 reports the annualized mean return, volatility and Sharpe ratio, and Figure 3.1 shows the cumulative returns.

Portfolio	Currency-hedged			Unhedged		
	Mean	Vol.	Sharpe	Mean	Vol.	Sharpe
<i>Six foreign indices</i>						
Equally Weighted	6.61%	13.32%	0.404	5.46%	17.54%	0.242
Risk Parity	6.52%	13.17%	0.402	5.33%	17.07%	0.240
Mean-Variance ($\gamma = 10$)	0.11%	16.47%	-0.068	2.74%	18.50%	0.082
Mean-Variance ($\gamma = 1$)	-9.91%	164.69%	-0.068	16.35%	185.11%	0.082
<i>US index + six foreign indices</i>						
Equally Weighted	7.15%	13.53%	0.438	—	—	—
Risk Parity	7.06%	13.39%	0.436	—	—	—
Mean-Variance ($\gamma = 10$)	1.41%	19.06%	0.010	—	—	—
US market alone (CRSP)	10.41%	16.42%	0.559	10.41%	16.42%	0.559

TABLE 2: Annualized performance of the portfolios (2007-05 to 2024-12). The US + foreign portfolios use the currency-hedged returns of the foreign indices.

With $\gamma = 1$, the mean-variance weights are ten times larger than with $\gamma = 10$. The portfolio is so leveraged that it loses more than 100% in a single month (-217% in January 2008 with hedged returns), which means the investor is wiped out. We therefore use $\gamma = 10$. The Sharpe ratio is unchanged, because γ only scales the risky positions.

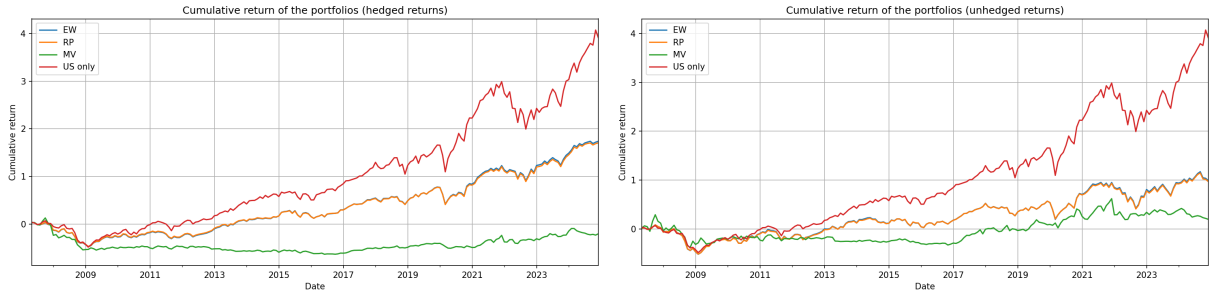


FIG. 3.1: Cumulative returns of the three portfolios ($\gamma = 10$ for the mean-variance portfolio) and of the US market, with currency-hedged (left) and unhedged (right) returns

Conclusion: Table 2 leads to five observations.

- **Hedging improves risk-adjusted performance.** The Sharpe ratio of the equally weighted and risk parity portfolios rises from about 0.24 to 0.40. This comes mainly from a much lower volatility (13% instead of 17–18%): currency risk adds volatility without being rewarded. The hedged portfolios also have a higher mean return, because the USD appreciated over the period, which hurt unhedged investors.
- **Equal weights and risk parity give almost identical results.** The six indices have similar volatilities, so inverse-volatility weights stay close to $1/N$.
- **Mean-variance performs worst**, with both the lowest mean and the highest volatility. This illustrates the limits of unconstrained mean-variance optimization documented by Black and Litterman (1991). Expected returns are estimated very imprecisely on 60 months, and the inverse of the covariance matrix amplifies these errors into extreme and unstable weights.
- **No portfolio beats the US market alone**, which has a Sharpe ratio of 0.56 over the same period. US equities strongly outperformed the rest of the world after 2009 (Figure 3.1).
- **Adding the US index does not change this result.** It raises the Sharpe ratio of the equally weighted and risk parity portfolios from 0.40 to 0.44, and that of the mean-variance portfolio from -0.07 to 0.01 , but all of them remain below the US market alone. With seven indices of similar volatility, the US index only receives about 14% of the weight. More generally, an asset B improves the Sharpe ratio of a portfolio A only if $SR_B > \rho_{A,B} SR_A$. With a correlation of 0.88 with the US market, the hedged risk parity portfolio of foreign indices would need a Sharpe ratio above $0.88 \times 0.559 = 0.49$, against 0.40.

Over 2007–2024, investing internationally therefore mainly reduced risk: the hedged portfolios have a volatility of about 13% against 16% for the US market. It did not improve the Sharpe ratio compared with holding US equities, even when foreign equities are added to US equities rather than replacing them. For a US investor, the benefit of international diversification is a lower volatility rather than a higher risk-adjusted return, and it is best obtained with currency-hedged positions. This result is specific to the period: US equities strongly outperformed after 2009, which an investor could not have known in 2007.

- (d) In the rest of the report, DIV denotes the currency-hedged risk parity portfolio of the six foreign indices, with return R^{DIV} . DIV does not include the US index, since it represents the international diversification strategy; the US market is only used as a benchmark. In the regressions, we use its excess return $R^{DIV} - R^{T-Bill}$, since DIV is a \$1 long position.

4 Equity Index Momentum Strategy (MOM)

- (a) Momentum relies on the idea that markets that performed well over the past year keep outperforming in the short run, because investors under-react to news. Every month t , we compute the momentum of each currency-hedged index as its cumulative return from $t - 12$ to $t - 2$, i.e. the 11-month return lagged by one month:

$$\text{MOM}_t^i = \prod_{\tau=t-12}^{t-2} (R_\tau^i + 1) - 1$$

The most recent month is skipped because returns tend to reverse at a one-month horizon. We rank the countries so that $\text{Rank}_t^i = 1$ for the country with the lowest momentum, and assign the weights

$$\omega_t^i = Z \left(\text{Rank}_t^i - \frac{N+1}{2} \right), \quad Z = \left(\sum_{k > \frac{N+1}{2}}^N \left(k - \frac{N+1}{2} \right) \right)^{-1}$$

The factor Z ensures that the positive weights sum to +1 and the negative weights to -1 (with $N = 6$, $Z = 1/4.5$). The strategy is therefore long past winners, short past losers, and costs nothing. Its return is $R_t^{\text{MOM}} = \sum_{i=1}^N \omega_t^i R_t^{i, \text{hedged}}$. The first signal is available after 12 months, so the strategy runs from 2003-05 to 2024-12 (260 months).

- (b) Table 3 reports the performance of the strategy and of its two legs. The long leg only keeps the positive weights and the short leg only the negative ones. Since the long-short strategy costs nothing, its return is already an excess return. For the legs, the Sharpe ratio uses the excess return over the T-Bill: $R - R^f$ for the long leg, and $R + R^f$ for the short leg, because the proceeds of a short sale earn the risk-free rate.

Portfolio Strategy	Mean	Standard Deviation	Sharpe Ratio
MOM Long Leg	7.82%	13.94%	0.454
MOM Short Leg	-9.63%	12.98%	-0.626
MOM Strategy	-1.80%	8.68%	-0.208

TABLE 3: Performance of the Momentum Strategy Components (2003-05 to 2024-12)

The cumulative returns are displayed on Figure 4.1. Each leg is dominated by the direction of the equity market. The long leg rises with the markets, and the short leg loses because it is short markets that went up on average. What matters is the difference between the two legs: the strategy drifts slowly downward, so past winners did not outperform past losers among these six markets.

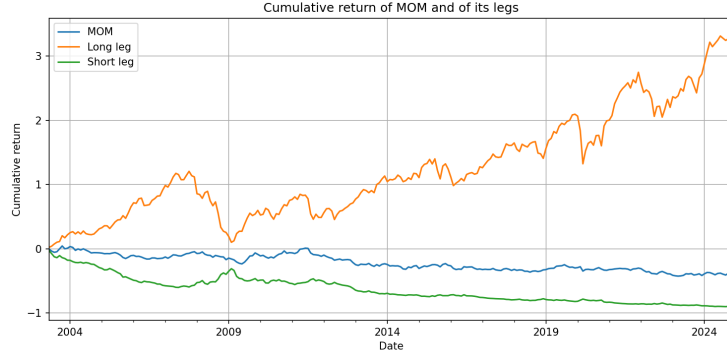


FIG. 4.1: Cumulative returns of the momentum strategy and of its legs

We then test whether the average return of MOM is statistically different from zero, with a one-sample t-test under the null hypothesis that the average return is 0. The t-statistic is -0.97 (p-value of 0.334). We cannot reject the null hypothesis at the 5% level: the average return of MOM is not significantly different from zero. The negative point estimate suggests that country momentum, if anything, reversed over the sample. The inverted strategy would earn $+1.80\%$ per year, but with the same absolute t-statistic, so this is not significant either.

(c) We regress the MOM return on the excess return of DIV by ordinary least squares:

$$R_t^{\text{MOM}} = \alpha + \beta (R_t^{\text{DIV}} - R_t^{T\text{-Bill}}) + \varepsilon_t$$

The results are:

- The intercept is $\alpha = -0.0018$ (-2.12% per year) and is not significant ($t = -0.99$, $p = 0.321$).
- The slope is $\beta = 0.076$ and is not significant at the 10% level ($t = 1.63$, $p = 0.105$).
- The R^2 is 0.012: DIV explains only 1.2% of the variation of MOM.

Conclusion: MOM is almost uncorrelated with DIV. A low correlation is not enough to make a strategy attractive, however. Adding MOM to a DIV portfolio improves the Sharpe ratio only if MOM has a non-zero alpha with respect to DIV, i.e. a return that DIV does not explain. Here the alpha is negative and not significant, so there is no evidence that a DIV investor would benefit from investing in MOM.

5 Equity Index Long Term Reversal strategy (REV)

- (a) Long-term reversal relies on the idea that investors over-react over several years. Markets that fell a lot become cheap and tend to rebound, while markets that rose a lot become expensive. This works like a value signal. Every month t , we compute the cumulative currency-hedged return of each index from $t - 60$ to $t - 13$, i.e. the 5-year return lagged by 12 months:

$$\text{REV}_t^i = \prod_{\tau=t-60}^{t-13} (R_\tau^i + 1) - 1$$

The last 12 months are excluded so that the signal does not overlap with momentum. We rank the countries so that $\text{Rank}_t^i = 1$ for the country with the lowest past return, and assign the weights

$$\omega_t^i = Z \left(\frac{N+1}{2} - \text{Rank}_t^i \right), \quad Z = \left(\sum_{k > \frac{N+1}{2}}^N \left(k - \frac{N+1}{2} \right) \right)^{-1}$$

so that the long positions sum to +1 and the short positions to -1. The weights are the opposite of momentum: the strategy is long past losers and short past winners. Its return is $R_t^{\text{REV}} = \sum_{i=1}^N \omega_t^i R_t^{i, \text{hedged}}$, from 2007-05 to 2024-12 (212 months).

- (b) Table 4 reports the mean, standard deviation and Sharpe ratio of the REV strategy and of its long and short legs. The Sharpe ratios follow the same conventions as for MOM.

Portfolio Strategy	Mean	Standard Deviation	Sharpe Ratio
REV Long Leg	6.73%	13.42%	0.411
REV Short Leg	-6.13%	14.27%	-0.344
REV Strategy	0.61%	7.79%	0.078

TABLE 4: Performance of the Reversal Strategy Components (2007-05 to 2024-12)

The cumulative returns are displayed on Figure 5.1. The short leg gains strongly during the 2008 crisis, because it is short falling markets, and then declines steadily as markets recover. The long leg is roughly its mirror image. The long-short strategy stays close to zero over the whole period.

We then test whether the average return of REV is statistically different from zero, with a one-sample t-test under the null hypothesis that the average return is 0. The t-statistic is 0.33 (p-value of 0.744). We cannot reject the null hypothesis at the 5% level: the average return of REV is not significantly different from zero.

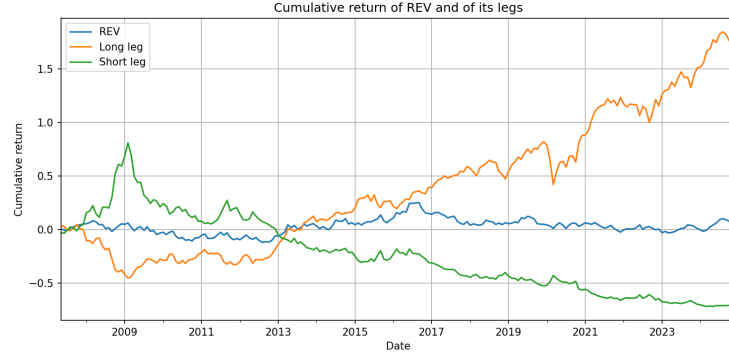


FIG. 5.1: Cumulative returns of the reversal strategy and of its legs

(c) We regress the REV return on the excess return of DIV by ordinary least squares, $R_t^{REV} = \alpha + \beta(R_t^{DIV} - R_t^{T-Bill}) + \varepsilon_t$. The results are:

- The intercept is $\alpha = 0.0009$ (+1.03% per year) and is not significant ($t = 0.55$, $p = 0.580$).
- The slope is $\beta = -0.080$ and is borderline significant at the 5% level ($t = -1.98$, $p = 0.050$).
- The R^2 is 0.018: DIV explains only 1.8% of the variation of REV.

Conclusion: REV is almost independent from DIV, with a slightly negative exposure. It tends to do a bit better when DIV falls, which gives it some diversification value. Its alpha is positive, but far from significant. At best, REV is a weak diversifier, and there is no statistical evidence that it would improve the portfolio of a DIV investor.

6 Currency Carry Strategy (CARRY)

- (a) The carry trade borrows in low-interest-rate currencies and invests in high-interest-rate currencies to earn the rate differential. Uncovered interest parity predicts that high-rate currencies should depreciate by exactly this differential. Empirically they do not fully, which creates a carry premium. The cost is a crash risk: high-rate currencies fall sharply in crises, when investors fly to safe-haven funding currencies such as the JPY and the CHF.

Every month, we sort the currencies on their interest rate differential $r_{t-1}^i - r_{t-1}^{US}$. This is the 3-month rate known at the start of the month, which is exactly the rate earned in X_t^i . Sorting on the differential is equivalent to sorting on the foreign rate. Since France and Germany share the euro and the same interbank rate, the euro is counted once, so we have $N = 5$ currencies (AUD, EUR, CHF, JPY, GBP). With $\text{Rank}_t^i = 1$ for the lowest differential, the weights are

$$\omega_t^i = Z \left(\text{Rank}_t^i - \frac{N+1}{2} \right), \quad Z = \left(\sum_{k > \frac{N+1}{2}}^N \left(k - \frac{N+1}{2} \right) \right)^{-1}$$

so that the positive weights sum to +1 and the negative weights to -1 (with $N = 5$, $Z = 1/3$, and the middle currency gets a zero weight). The return of the strategy is

$$R_t^{CARRY} = \sum_{i=1}^N \omega_t^i X_t^i$$

It runs from 2002-05 to 2024-12 (272 months).

- (b) Table 5 reports the mean, standard deviation and Sharpe ratio of the CARRY strategy and of its long and short legs. The currency returns X^i already include the cost of borrowing in USD, so they are excess returns. The Sharpe ratios of the strategy and of both legs are therefore computed without subtracting the risk-free rate.

Portfolio Strategy	Mean	Standard Deviation	Sharpe Ratio
CARRY Long Leg	2.02%	10.07%	0.201
CARRY Short Leg	-0.27%	8.32%	-0.032
CARRY Strategy	1.76%	9.09%	0.193

TABLE 5: Performance of the Currency Carry Strategy Components (2002-05 to 2024-12)

The cumulative returns are plotted on Figure 6.1, which shows the three phases of the carry trade:

- **Until mid-2008**, the strategy and its long leg rise steadily.
- **In the 2008 crisis**, they collapse: the high-rate currencies (e.g. the AUD) crash while the funding currencies (JPY, CHF) appreciate. This is the typical carry crash, and it wipes out most of the gains accumulated since 2002.
- **After 2009**, the long leg recovers until 2011–2013 (peaking above +100%) and then declines, while the long-short strategy stays roughly flat.

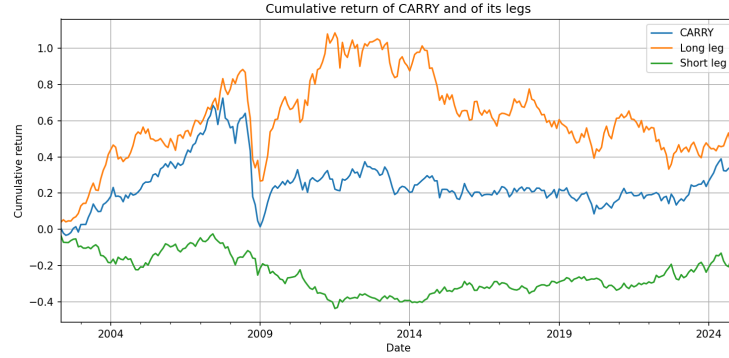


FIG. 6.1: Cumulative returns of the carry strategy and of its legs

We then test whether the average return of CARRY is statistically different from zero, with a one-sample t-test under the null hypothesis that the average return is 0. The t-statistic is 0.92 (p-value of 0.358). We cannot reject the null hypothesis at the 5% level: the average return of CARRY is not significantly different from zero.

- (c) We regress the CARRY return on the excess return of DIV by ordinary least squares, $R_t^{CARRY} = \alpha + \beta(R_t^{DIV} - R_t^{T-Bill}) + \varepsilon_t$. The results are:
- The intercept is $\alpha = -0.0021$ (−2.54% per year) and is not significant ($t = -1.31$, $p = 0.192$).
 - The slope is $\beta = 0.385$ and is highly significant ($t = 9.10$, $p < 0.001$). When the excess return of DIV increases by 1%, the CARRY return increases by about 0.39% on average.
 - The R^2 is 0.283: DIV explains 28.3% of the variation of CARRY, much more than for MOM and REV.

Conclusion: CARRY is a “risk-on” strategy. It is exposed to the same risk as equities and loses in crises, precisely when DIV loses. Once this exposure is accounted for, its alpha is negative and not significant. Adding CARRY to a DIV portfolio would therefore add risk of the same kind without any extra reward, so it is not attractive for a DIV investor.

7 Currency dollar Strategy (DOLLAR)

- (a) The assignment describes a long-dollar strategy against the rest of the world, i.e. a portfolio that goes short an equal-weighted basket of the $N = 5$ foreign currencies (the euro counted once). Its return is

$$R_t^{\text{DOLLAR}} = -\frac{1}{N} \sum_{i=1}^N X_t^i$$

The formula given in the assignment, $+\frac{1}{N} \sum_i X_t^i$, corresponds to the opposite position (long the foreign currencies). We follow the verbal description; with the other convention, only the signs of the results below change. Intuitively, the USD is a safe-haven currency that tends to appreciate in crises, so this strategy should hedge international equity exposure.

- (b) Table 6 reports the performance of the DOLLAR strategy. As for CARRY, the X^i are excess returns, so the Sharpe ratio does not subtract the risk-free rate.

Portfolio Strategy	Mean	Standard Deviation	Sharpe Ratio
DOLLAR Strategy	-0.62%	7.54%	-0.082

TABLE 6: Performance of the Currency Dollar Strategy (2002-05 to 2024-12)

We then test whether the average return of DOLLAR is statistically different from zero, with a one-sample t-test under the null hypothesis that the average return is 0. The t-statistic is -0.39 (p-value of 0.697). We cannot reject the null hypothesis at the 5% level: the average return of DOLLAR is not significantly different from zero.

The cumulative return is displayed on Figure 7.1. The strategy loses until 2011, while the USD depreciates against the other currencies. It then recovers as the USD strengthens, in particular after 2014 and in 2022.

- (c) We regress the DOLLAR return on the excess return of DIV by ordinary least squares, $R_t^{\text{DOLLAR}} = \alpha + \beta(R_t^{\text{DIV}} - R_t^{\text{T-Bill}}) + \varepsilon_t$. The results are:
- The intercept is $\alpha = 0.0017$ (+2.06% per year) and is not significant ($t = 1.19$, $p = 0.235$).
 - The slope is $\beta = -0.194$ and is highly significant ($t = -5.17$, $p < 0.001$). This shows a significant *negative* relationship between DOLLAR and DIV.
 - The R^2 is 0.113: DIV explains 11.3% of the variation of DOLLAR.

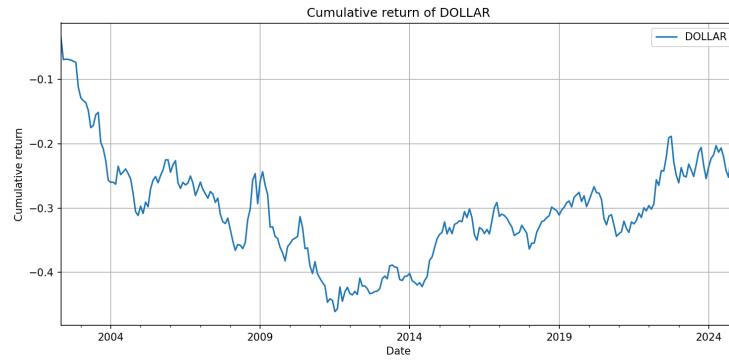


FIG. 7.1: Cumulative return of the DOLLAR strategy

Conclusion: DOLLAR tends to gain when DIV loses, and conversely. This is consistent with the safe-haven role of the USD: when foreign equities fall, investors buy dollars. For a DIV investor, DOLLAR does not add exposure to the same drivers of return; instead, it hedges the periods in which DIV declines. Its alpha is positive, which makes it the most promising of the four strategies as a complement to DIV, but the alpha is not statistically significant.

8 Optimal Fund Portfolio Return (STRAT)

1. A fund investing in the 1-month T-Bill and in DIV has a return of the form $R_{FUND} = R_{T-Bill} + a(R_{DIV} - R_{T-Bill})$. We target an annualized volatility of $\sigma_{FUND} = 15\%$. The T-Bill return is known at the beginning of each month, so it does not generate any volatility, and the volatility of the fund is $a \times \sigma(R_{DIV} - R_{T-Bill})$.

- **Full-sample value.** If a is chosen so that the volatility over the whole history is exactly 15%, we obtain $a = 1.14$.
- **Rolling value.** This full-sample value uses future information. For the fund, we therefore compute a on a rolling window of 60 months. Let σ_{DIV}^t be the standard deviation of $R_{DIV} - R_{T-Bill}$ over $[t - 59, t]$. The weight used from t to $t + 1$ is

$$a^t = \frac{\sigma_{FUND}}{\sqrt{12} \times \sigma_{DIV}^t}$$

The time series of a^t is shown on Figure 8.1. It varies between 0.88 and 1.67, with an average of $\bar{a} = 1.27$. It moves inversely to the estimated volatility of DIV:

- it stays around 0.9 while the window contains the 2008 crisis;
- it rises when the crisis leaves the window at the end of 2013, and peaks around 1.6 in 2017–2018;
- it drops in 2020, when the Covid shock enters the window.

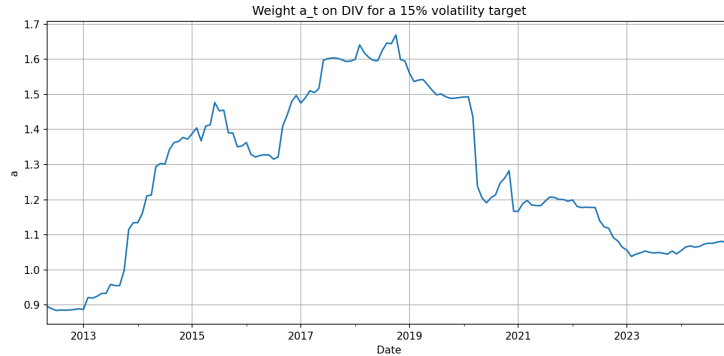


FIG. 8.1: Rolling weight a^t on DIV to reach an annualized volatility of 15%

2. We then combine the four strategies MOM, REV, CARRY and DOLLAR with risk parity weights, computed exactly as for DIV: inverse volatility over the past 60 months, normalized to sum to 1. We call R_{STRAT} the return of this combination. Since the four strategies cost nothing, STRAT is also a zero-cost overlay. REV only starts in 2007-05 and the risk parity weights need 60 months of history, so STRAT runs from 2012-05 to 2024-12 (152 months).

The cumulative return is displayed on Figure 8.2 and the weights on Figure 8.3. The weights remain between 0.18 and 0.32. DOLLAR, the least volatile strategy, receives the largest weight during most of the period (0.29 on average), while MOM receives the lowest weight after 2014. The weight of CARRY jumps at the end of 2013, when the 2008 carry crash leaves the 60-month window and its estimated volatility falls. The performance of STRAT is given in Table 7: its mean return is small but positive, with a low volatility.

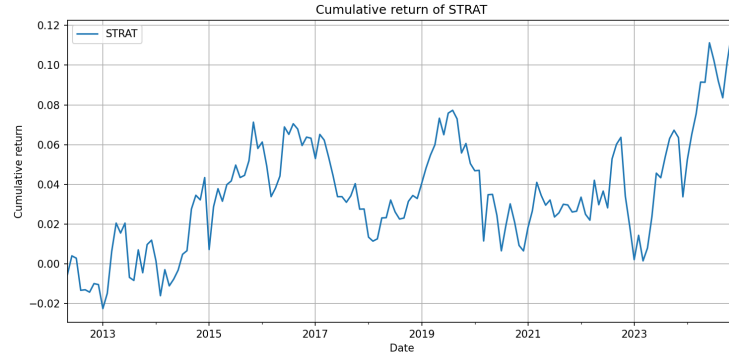


FIG. 8.2: Cumulative return of STRAT

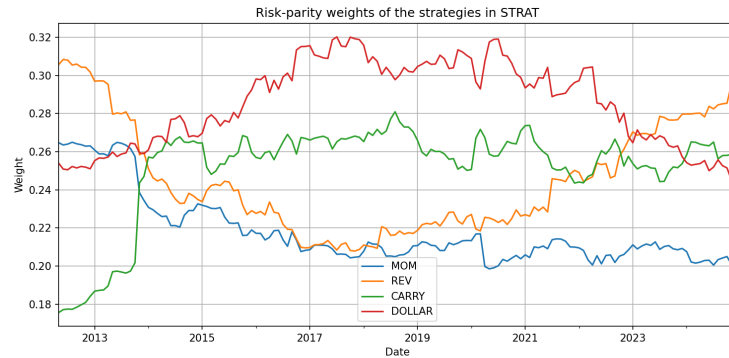


FIG. 8.3: Risk parity weights of the four strategies in STRAT

Portfolio Strategy	Mean	Standard Deviation	Sharpe Ratio
STRAT Strategy	0.92%	3.91%	0.234

TABLE 7: Performance of the STRAT rolling risk parity strategy (2012-05 to 2024-12)

3. The new fund has the return $R_{FUND} = R_{T-Bill} + b \times (R_{DIV} - R_{T-Bill}) + c \times R_{STRAT}$, still with an annualized volatility target of $\sigma_{FUND} = 15\%$. The only sources of risk

are the two excess returns $R_{DIV} - R_{T-Bill}$ and R_{STRAT} . Let Σ_t be their covariance matrix and μ_t their vector of mean returns, both estimated over $[t - 59, t]$. On every rolling window, we solve

$$\max_{w_t} w_t^\top \mu_t \quad \text{such that} \quad \sqrt{12 \times w_t^\top \Sigma_t w_t} = \sigma_{FUND}$$

Since the objective is linear and the constraint is quadratic, the solution is proportional to $\Sigma_t^{-1} \mu_t$ and scaled to meet the volatility target:

$$w_t = \begin{pmatrix} b_t \\ c_t \end{pmatrix} = \frac{\sigma_{FUND} \times \Sigma_t^{-1} \mu_t}{\sqrt{12 \times \mu_t^\top \Sigma_t^{-1} \mu_t}}$$

where b_t and c_t are the weights on $R_{DIV} - R_{T-Bill}$ and R_{STRAT} from t to $t + 1$. Since STRAT only starts in 2012-05 and (b_t, c_t) need 60 months of STRAT returns, the fund runs from 2017-05 to 2024-12 (92 months).

The time series of b_t and c_t are displayed on Figure 8.4. On average, $\bar{b} = 1.27$ and $\bar{c} = -0.16$.

- b_t is stable, between 0.96 and 1.66.
- c_t is very unstable, between -3.75 and $+2.49$. It is positive until 2019, after a good run of STRAT. It collapses to -3.7 in April 2020, after the Covid shock, and stays mostly negative until mid-2023. It is positive again afterwards.

This instability comes from the estimation of the mean return of STRAT. This mean is small and noisy, and the optimal weight is proportional to it, which is the same estimation problem as for the mean-variance portfolio of Section 3.

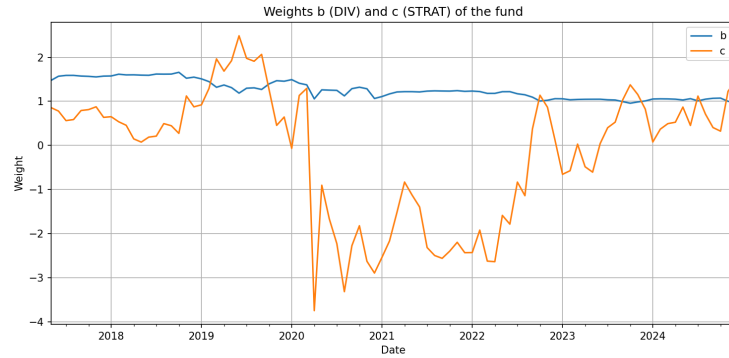


FIG. 8.4: Evolution of b_t and c_t when targeting $\sigma_{FUND} = 15\%$ on rolling windows

The time series of the weights invested in each asset and strategy are displayed on Figure 8.5, and their averages are reported in Table 8.

- **T-Bill and DIV.** The fund invests b_t in DIV and $1 - b_t$ in the T-Bill. The T-Bill weight is mostly negative: the fund borrows at the risk-free rate to lever DIV up to the 15% volatility target.

- **Sub-strategies.** Since STRAT is a zero-cost overlay, each of MOM, REV, CARRY and DOLLAR receives c_t times its risk parity weight in STRAT. The four weights therefore move together with c_t : positive before 2020, strongly negative from 2020 to 2022, and positive again at the end of the sample.

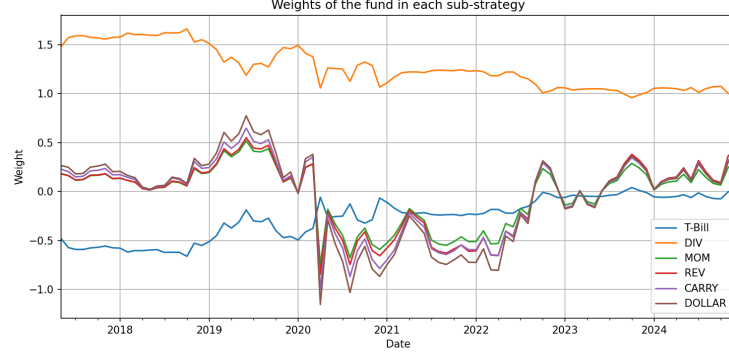


FIG. 8.5: Weights of the fund in each asset and sub-strategy

Substrategy	Average Optimal Weight
T-Bill	−0.265
DIV	1.265
MOM	−0.031
REV	−0.037
CARRY	−0.039
DOLLAR	−0.050

TABLE 8: Average weights of the fund in each asset and sub-strategy (2017-05 to 2024-12)

4. We compare the two funds over the same period (2017-05 to 2024-12):
 - the simple fund, invested in the T-Bill and DIV only, with the rolling weight a^t ;
 - the overlay fund, which also invests in STRAT.

Both target a volatility of 15%. Their performance is given in Table 9, and Figure 8.6 plots the value of \$1 invested in each fund.

The simple fund clearly outperforms: it has both a higher mean return and a lower volatility, and its Sharpe ratio is 0.53 against 0.32. The two funds are almost identical until 2020. The gap opens after the Covid shock: c_t turned strongly negative, based on the past performance of STRAT, just as STRAT recovered. The rolling mean-variance allocation thus chased noisy estimates of the mean of STRAT. Both funds

also have a realized volatility above the 15% target, because the volatility is estimated on past data and 2020 brought a volatility spike that the past windows did not anticipate.

Strategy	Mean	Standard Deviation	Sharpe Ratio
T-Bill + DIV (Simple Fund)	10.68%	16.18%	0.527
Overlay Strategy Fund	7.62%	17.20%	0.318

TABLE 9: Performance comparison of the simple and overlay funds (2017-05 to 2024-12)

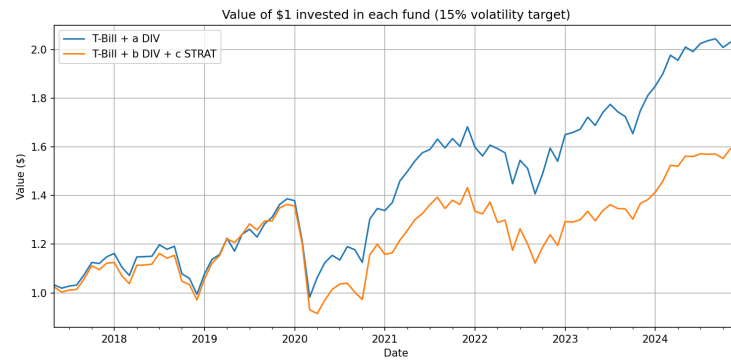


FIG. 8.6: Value of \$1 invested in each fund

9 Performance and risk analysis for the Fund strategy

- a) We regress the excess return of the overlay fund on the Fama-French 5 research factors:

$$R_t^{FUND} - RF_t = \alpha + \beta_{MKT} (R_t^M - RF_t) + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \varepsilon_t$$

RF is the risk-free rate. It is used to compute the excess return of the fund and is not a regressor. The sample covers the 92 months of the fund (2017-05 to 2024-12). The results are reported in Table 10.

Factor	Coefficient	t-stat
α (monthly)	-0.0034	-1.01
Mkt-RF	0.779	10.66
SMB	0.006	0.04
HML	0.207	1.68
RMW	0.051	0.30
CMA	-0.046	-0.26
R^2 (adjusted)	0.644 (0.624)	

TABLE 10: Fama-French 5-factor regression of the fund excess return (2017-05 to 2024-12)

Only the market factor has a t-statistic above 1.96 in absolute value: unconditionally, the fund loads significantly on the US market, with a beta of 0.78. HML is only significant at the 10% level ($t = 1.68$). This reflects a mild value tilt, consistent with foreign markets being more value-oriented than the US. SMB, RMW and CMA are not significant.

The R^2 is 0.644: the five US factors explain about two thirds of the variation of the fund. The remaining third comes from risks these purely US-based factors do not capture, namely country-specific equity risk and the currency strategies (CARRY and DOLLAR are driven by exchange rates, not by US equity factors). The alpha is -0.34% per month (-4.05% per year) and is not significant ($t = -1.01$): we cannot reject $\alpha = 0$. Overall, the US factors explain the equity part of the fund well, mostly through the market, but not its international and currency components.

- b) We discuss how these results relate to three asset pricing theories. The fund only covers 92 months, so all tests have limited power.
- **Efficient markets:** In an efficient market, predictable patterns should not generate abnormal returns once risk is accounted for. We find no significant

alpha for the fund. None of the four dynamic strategies (MOM, REV, CARRY, DOLLAR) has an average return significantly different from zero either, and none has a significant alpha with respect to DIV. The past-return and interest-rate signals therefore did not produce a “free lunch” in these markets, which is consistent with the weak form of market efficiency.

- **CAPM:** The market beta is the only significant coefficient, and the alpha is not significantly different from zero. This is in line with the CAPM, which states that expected returns are explained by exposure to the market alone. However, a third of the variance of the fund is not explained by the US market. Under the CAPM, this non-market risk should not be rewarded, which is consistent with the absence of a significant alpha.
- **APT:** The APT allows several risk factors to be priced. Among the five factors, only the market is significant and HML is marginally significant. The unexplained third of the variance suggests that relevant factors are missing, for instance international equity factors or currency factors such as the carry and dollar factors of Lustig, Roussanov and Verdelhan (2011). Our results therefore do not contradict the APT, but they show that purely US-based factors are not sufficient to explain an international equity and currency strategy.